Rahul Lohra

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Professional Summary

Results-driven ML/AI Engineer and Full-Stack Developer with hands-on experience in building production-grade machine learning pipelines, federated learning systems, and enterprise web applications. Proficient in Python-based ML frameworks, MERN stack development, and RESTful API design. Completed MCA from BHU; delivered security-critical systems at CERT-In serving 100+ users.

Education

Banaras Hindu University (BHU) **Jun 2025**
Master of Computer Applications (MCA), CGPA: 7.31/10 *Varanasi, India*

Dr. Shyama Prasad Mukherjee University **Aug 2023**
Bachelor of Science in Information Technology, CGPA: 8.61/10 *Ranchi, India*

Professional Experience

Intern – Full Stack & Security **Feb 2025 – Aug 2025**
CERT-In (Indian Computer Emergency Response Team) *Delhi, India*

- Architected and deployed a full-stack Django web application for 100+ internal users, improving workflow efficiency by 40% through automated document processing and RBAC.
- Designed robust RESTful API solutions using Twilio and Gmail for real-time authentication; applied async programming (Celery + Redis), reducing manual processing time by 60%.
- Implemented enterprise-grade security using PBKDF2-SHA256 hashing, MFA (TOTP/OTP), session management, and brute-force protection — achieving zero security incidents.
- Containerised services with Docker and configured Nginx reverse proxy; collaborated in Agile/Scrum sprints with 8+ engineers and maintained Swagger/OpenAPI documentation.

Projects

AI-Generated Image / Video Detection System | GitHub **Jan 2024 – Apr 2024**

- Fine-tuned ResNet on 80K samples; achieved 94.3% accuracy. Engineered frequency-domain feature pipeline (FFT + DCT) boosting recall by 18%.
- Built video analysis module using OpenCV; deployed as Flask REST API with React.js dashboard for drag-and-drop media upload and anomaly visualisation.

Federated Learning System for Medical Insurance **Mar 2024 – Aug 2024**

- Simulated privacy-preserving FL framework across 5 virtual clients; implemented FedAvg achieving $R^2 = 0.85$ without raw data sharing.
- Built modular Python ML pipeline with pytest, MLflow experiment tracking, and reproducible Jupyter notebooks.

Sentiment Analysis API & Dashboard **Sep 2023 – Feb 2024**

- Fine-tuned BERT classifier on 50K reviews achieving 92% F1-score; served via FastAPI with async inference, rate-limiting, and API-key authentication.

AI-Powered File Converter & Summarizer **Nov 2024 – Feb 2025**

- Built MERN-stack platform for multi-format conversion and AI summarisation using Hugging Face Transformers; deployed on AWS EC2 with S3 and CloudFront CDN.

Technical Skills

ML / AI: Python, Scikit-learn, PyTorch, Hugging Face Transformers, BERT, BART, ResNet, EfficientNet
Data Engineering: NumPy, Pandas, Matplotlib, Seaborn, SQL, MongoDB, Feature Engineering, EDA
Full-Stack Web: React.js, Node.js, Express.js, Django, FastAPI, HTML5/CSS3, REST APIs
DevOps & Cloud: Docker, Git/GitHub, AWS, Nginx, Linux, CI/CD
Security: MFA/OTP, PBKDF2-SHA256, RBAC, OWASP Top 10, Vulnerability Assessment

Certifications

Machine Learning Specialization, Deep Learning Specialization, Convolutional Neural Networks | *Coursera / DeepLear*